

HEALTHCARE DATA CLIENT

Data Dictionary

Healthcare Data Quality Automation and Exception Reporting

Document Control	Detail
Engagement	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Data Domains	Claims, patients, appointments, providers, billing, transactions, clinical records and insurance data
Sources	Excel, CSV and SQL/database
Batch Scale	Approximately 250,000 records per batch
Dictionary boundary: Exact historical column names, database data types, lengths, keys and source schemas are not retained. This dictionary therefore defines the analytical field classes required by the confirmed validation scope, without inventing client schema.	

1. Core Analytical Field Classes

Field Class	Business Meaning	Typical Analytical Type	Validation Relevance
Record Identifier	Identifier used to distinguish and trace a source record.	Text / identifier	Invalid ID, duplicate detection, traceability
Source / File Reference	Identifies the source file, extract or batch context.	Text	Reconciliation and lineage
Healthcare Entity Identifier	Identifier associated with a patient, claim, provider, appointment or other healthcare entity where present.	Text / identifier	Invalid ID, format and cross-field validation
Date Field	Date associated with a healthcare record or event.	Date / datetime	Invalid date and cross-field validation
Code Field	Coded healthcare or operational value present in the source.	Text / code	Inconsistent-code validation
Numeric Amount / Value	Numeric value such as an amount, count or measurable source value where present.	Decimal / integer	Range and format validation
Category / Status Field	Categorical source value used to describe a record state or type.	Text / category	Consistency and cross-field validation
Free-Text Field	Descriptive text where present in the source.	Text	Completeness or format checks where defined

2. Healthcare Domain Coverage

Domain	Confirmed Presence	Schema Treatment
Claims	Yes	Exact claim fields are not retained and are not reconstructed as historical columns.
Patients	Yes	Exact patient fields and identifiers are not exposed.
Appointments	Yes	Exact appointment fields are not retained.
Providers	Yes	Exact provider fields are not retained.
Billing	Yes	Exact billing fields are not retained.
Transactions	Yes	Exact transaction fields are not retained.
Clinical records	Yes	Exact clinical fields are not retained.
Insurance data	Yes	Exact insurance fields are not retained.

Confidentiality: No patient names, member IDs, claim numbers, clinical values, provider identifiers or other confidential healthcare values are introduced into this portfolio dictionary.

3. Validation Metadata Fields

Portfolio Field	Definition	Type	Status
Validation_Status	Indicates whether the processed record passed applicable validation or requires review.	Category	Reconstructed implementation field
Exception_Flag	Record-level indicator showing whether one or more applicable rules failed.	Boolean / flag	Reconstructed implementation field
Reason_Code	Code or label identifying the validation failure associated with an exception.	Text / code	Confirmed concept; exact historical vocabulary not retained
Validation_Rule	Identifies the rule that generated a validation failure.	Text	Reconstructed implementation field
Validation_Message	Readable explanation of the validation failure.	Text	Reconstructed supporting field
Source_Record_Reference	Traceable reference back to the processed source record.	Text / identifier	Reconstructed traceability field
Batch_Reference	Identifies the processed batch or file context.	Text / identifier	Reconstructed supporting field

4. Data Quality Rule Mapping

Quality Check	Required Field Characteristic	Validation Intent
Duplicate check	Record identifier and/or documented duplicate-defining fields	Identify records meeting the implemented duplicate rule.
Null check	Field identified as subject to completeness validation	Identify missing values where completeness is required.
Invalid date	Date field	Identify values that cannot be interpreted under the documented date rule.
Incorrect format	Field with a documented expected format	Identify values not matching the required structure.
Invalid ID	Identifier field	Identify identifiers failing the documented validity rule.
Out-of-range value	Numeric or date field with documented valid bounds	Identify values outside the defined range.
Inconsistent code	Code or category field	Identify values inconsistent with the documented valid code set or mapping.
Cross-field mismatch	Two or more related fields	Identify records where documented relationships between fields are not satisfied.

5. Field Governance Rules

Governance Area	Rule
Names	Use source field names in the implementation when known. Do not invent historical column names for portfolio evidence.
Data types	Profile and convert fields to the analytical type required for validation while retaining source traceability.
Nulls	Do not replace missing healthcare values with fabricated values solely to pass validation.
Identifiers	Treat identifiers as identifiers, not measures. Preserve leading characters where relevant to the source.
Dates	Standardize to a valid analytical date representation only after parsing and validation.
Codes	Use documented mappings or valid sets. Do not infer unretained healthcare code lists.
Exceptions	Retain the source reference and validation reason needed for targeted review.
Confidentiality	Exclude PHI and confidential client values from public portfolio artifacts.

6. Exception Data Structure

Logical Element	Purpose
Source record reference	Connect exception output to the affected processed record.
Exception flag	Distinguish records requiring review.
Reason code	Classify the validation failure.
Validation rule	Identify which check failed.
Validation message	Explain the failure to the reviewer.
Relevant source field/value	Provide review context in the controlled client environment where appropriate.
Batch reference	Support reconciliation to the processed file or batch.

7. Record-Level and Rule-Level Grain

Dataset / View	Recommended Grain	Purpose
Processed record view	One row per governed processed record	Supports Records Processed and record-level exception status.
Exception record view	One row per governed exception record	Supports Exception Records and record-level Exception Rate.
Validation failure view	One row per record per failed validation rule	Supports multiple failures and reason-code analysis without distorting Exception Rate.

Important modeling control: One record may fail multiple rules. Record-level Exception Rate must count the affected record once, while a rule-level failure view may contain multiple rows for that same record.

8. KPI Field Dependencies

KPI / Measure	Required Data Elements
Records Processed	Governed processed-record population
Exception Records	Exception flag or equivalent record-level validation result
Exception Rate	Records Processed and distinct Exception Records
Validation Failures	Rule-level validation results
Exceptions by Reason Code	Exception record reference plus reason code or rule
Review Turnaround Time	Review timing evidence at file level where retained
Review Time Reduction	Confirmed baseline and post-solution review times

9. Source-to-Reporting Lineage

Layer	Data Dictionary Role
Source	Retain original source fields and source reference where available.
Preparation	Profile types, formats, nulls and standardization requirements.
Validation	Apply documented checks to applicable fields.
Exception output	Add exception status, reason and traceability metadata.
Reporting	Use governed processed and exception populations for Power BI measures.

10. Data Dictionary Validation Checklist

Check	Expected Control
Field existence	Required source field exists before its rule is executed.
Type readiness	Field can be safely interpreted using the documented analytical type.
Null behavior	Missing values are handled according to the applicable validation rule.
Rule mapping	Each implemented validation rule maps to the correct source field or field combination.
Reason traceability	Each failure can be associated with its reason and source record.
KPI grain	Record-level measures are not inflated by rule-level duplicate rows.
Confidentiality	Public outputs contain no real patient or confidential healthcare values.

11. Portfolio Integrity

Use of this dictionary: This document deliberately avoids fabricated claim, patient, provider, billing, clinical and insurance field names. When the original schema is unavailable, professional data governance is better demonstrated by documenting field classes, validation dependencies, grain and traceability than by inventing a healthcare schema.